

# When does RSM beat both benchmarks?

The three losses.

$$L^{\text{ave}} = s + Q^{\text{ave}} \quad (\text{no inflation})$$

$$L_k^{\text{bag},\infty} = (A + R_k^{\text{bag},\infty}) \frac{T-1}{T-k}, \quad A = s + Q^{\text{opt}}$$

$$L^{\text{full}} = A \frac{T-1}{T-m} \quad (\text{full estimated optimal})$$

## 1. RSM beats the full estimated optimal.

Exact equivalence:

$$L_k^{\text{bag},\infty} < L^{\text{full}} \iff R_k^{\text{bag},\infty} < A \cdot \frac{m-k}{T-m}.$$

Certified sufficient condition, using the exact Jensen ceiling  $R_k^{\text{bag},\infty} \leq Q_k^{\text{sub}} - Q^{\text{opt}}$  from Proposition 7 (the exact bracket):

$$\boxed{Q_k^{\text{sub}} - Q^{\text{opt}} < A \cdot \frac{m-k}{T-m}}$$

Leading order ( $R_k^{\text{ceil}} \approx Bz_k$ ,  $z_k = \frac{m-k}{km}$ , valid for  $1 \ll k \ll m$ ) — the  $(m-k)$  factor cancels on both sides:

$$B \cdot \frac{m-k}{km} < A \cdot \frac{m-k}{T-m} \implies \boxed{k \cdot m > \frac{B(T-m)}{A}}$$

## 2. RSM beats equal weighting.

With  $\Delta^{\text{ave}} := Q^{\text{ave}} - Q^{\text{opt}} = \frac{B(H-1)}{m}$  (exact):

$$L_k^{\text{bag},\infty} < L^{\text{ave}} \iff (A + R_k^{\text{bag},\infty}) \frac{T-1}{T-k} < A + \Delta^{\text{ave}}.$$

Certified sufficient condition:

$$\boxed{(A + Q_k^{\text{sub}} - Q^{\text{opt}}) \frac{T-1}{T-k} < A + \frac{B(H-1)}{m}}$$

Leading order, further taking  $T \rightarrow \infty$  (so  $\frac{T-1}{T} \rightarrow 1$ ):

$$\boxed{k > \frac{m}{H-1}}$$

(Sanity check: as  $H \rightarrow 1$  this threshold  $\rightarrow \infty$  — no finite  $k$  beats equal weighting, matching Remark 7 (the flat-risk case)'s  $k^* = 1$  result exactly.)

**Caveat.** The leading-order boxes use  $Q_k^{\text{sub}} - Q^{\text{opt}} \approx Bz_k$ , which the paper itself only claims is accurate for  $1 \ll k \ll m$  (the average-subset approximation's own stated scope). At small  $k$  this drops Lemma 4 (Taylor expansion)'s correction term  $\text{Var}_S(A_S)/(\kappa\bar{\tau})^3$  and can understate the true ceiling. The two *exact* boxes (using  $Q_k^{\text{sub}}$  directly) do not have this problem and are valid at any  $k$ .